MILITARY SPECIFICATION SHEET

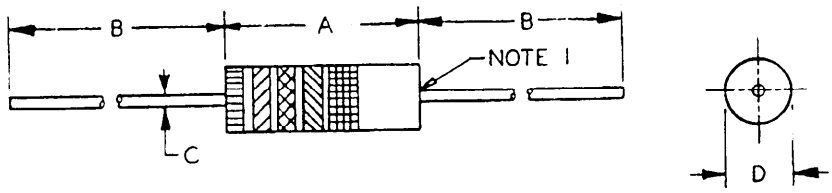
RESISTORS, FIXED, FILM, INSULATED,

STYLE RL32

Ⓑ INACTIVE FOR DESIGN AFTER DATE OF THIS SPECIFICATION SHEET (SEE PAGE 7). USE MIL-R-39017/3.

The complete requirements for procuring the resistors described herein shall consist of this document and the latest issue of Specification MIL-R-22684.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.



LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
A	.520	.593	13.21	15.06	
B	1.375	1.625	34.92	41.28	
C	.038	.042	.96	1.07	
D	.175	.205	4.45	5.21	

NOTES:
1. The end of the body shall be that point at which the diameter equals the nearest drill size larger than 250 percent of the nominal lead diameter. The leads shall be solderable to within .125 (3.18 mm) of the resistor body.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Resistance values: See table I.

Resistance tolerances: See table I.

Terminal type: See table I.

Minimum resistance value: 10 ohms.

Maximum resistance value: 1.0 megohm.

Ⓑ denotes changes

FSC 5905

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TABLE I. Part number designation

Type designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.
	Ohms			
RL32S100G	10	G	S	0001
RL32S100J	10	J	S	0002
RL32S110G	11	G	S	0003
RL32S110J	11	J	S	0004
RL32S120G	12	G	S	0005
RL32S120J	12	J	S	0006
RL32S130G	13	G	S	0007
RL32S130J	13	J	S	0008
RL32S150G	15	G	S	0009
RL32S150J	15	J	S	0010
RL32S160G	16	G	S	0011
RL32S160J	16	J	S	0012
RL32S180G	18	G	S	0013
RL32S180J	18	J	S	0014
RL32S200G	20	G	S	0015
RL32S200J	20	J	S	0016
RL32S220G	22	G	S	0017
RL32S220J	22	J	S	0018
RL32S240G	24	G	S	0019
RL32S240J	24	J	S	0020
RL32S270G	27	G	S	0021
RL32S270J	27	J	S	0022
RL32S300G	30	G	S	0023
RL32S300J	30	J	S	0024
RL32S330G	33	G	S	0025
RL32S330J	33	J	S	0026
RL32S360G	36	G	S	0027
RL32S360J	36	J	S	0028
RL32S390G	39	G	S	0029
RL32S390J	39	J	S	0030
RL32S430G	43	G	S	0031
RL32S430J	43	J	S	0032
RL32S470G	47	G	S	0033
RL32S470J	47	J	S	0034
RL32S510G	51	G	S	0035
RL32S510J	51	J	S	0036
RL32S560G	56	G	S	0037
RL32S560J	56	J	S	0038
RL32S620G	62	G	S	0039
RL32S620J	62	J	S	0040
RL32S680G	68	G	S	0041
RL32S680J	68	J	S	0042
RL32S750G	75	G	S	0043
RL32S750J	75	J	S	0044
RL32S820G	82	G	S	0045
RL32S820J	82	J	S	0046
RL32S910G	91	G	S	0047
RL32S910J	91	J	S	0048
RL32S101G	100	G	S	0049
RL32S101J	100	J	S	0050

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TABLE I. Part number designation - Continued

Type designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.
	Ohms			
RL32S111G	110	G	S	0051
RL32S111J	110	J	S	0052
RL32S121G	120	G	S	0053
RL32S121J	120	J	S	0054
RL32S131G	130	G	S	0055
RL32S131J	130	J	S	0056
RL32S151G	150	G	S	0057
RL32S151J	150	J	S	0058
RL32S161G	160	G	S	0059
RL32S161J	160	J	S	0060
RL32S181G	180	G	S	0061
RL32S181J	180	J	S	0062
RL32S201G	200	G	S	0063
RL32S201J	200	J	S	0064
RL32S221G	220	G	S	0065
RL32S221J	220	J	S	0066
RL32S241G	240	G	S	0067
RL32S241J	240	J	S	0068
RL32S271G	270	G	S	0069
RL32S271J	270	J	S	0070
RL32S301G	300	G	S	0071
RL32S301J	300	J	S	0072
RL32S331G	330	G	S	0073
RL32S331J	330	J	S	0074
RL32S361G	360	G	S	0075
RL32S361J	360	J	S	0076
RL32S391G	390	G	S	0077
RL32S391J	390	J	S	0078
RL32S431G	430	G	S	0079
RL32S431J	430	J	S	0080
RL32S471G	470	G	S	0081
RL32S471J	470	J	S	0082
RL32S511G	510	G	S	0083
RL32S511J	510	J	S	0084
RL32S561G	560	G	S	0085
RL32S561J	560	J	S	0086
RL32S621G	620	G	S	0087
RL32S621J	620	J	S	0088
RL32S681G	680	G	S	0089
RL32S681J	680	J	S	0090
RL32S751G	750	G	S	0091
RL32S751J	750	J	S	0092
RL32S821G	820	G	S	0093
RL32S821J	820	J	S	0094
RL32S911G	910	G	S	0095
RL32S911J	910	J	S	0096
RL32S102G	1,000	G	S	0097
RL32S102J	1,000	J	S	0098
RL32S112G	1,100	G	S	0099
RL32S112J	1,100	J	S	0100

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TABLE I. Part number designation - Continued

Type designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.
	Ohms			
RL32S122G	1,200	G	S	0101
RL32S122J	1,200	J	S	0102
RL32S132G	1,300	G	S	0103
RL32S132J	1,300	J	S	0104
RL32S152G	1,500	G	S	0105
RL32S152J	1,500	J	S	0106
RL32S162G	1,600	G	S	0107
RL32S162J	1,600	J	S	0108
RL32S182G	1,800	G	S	0109
RL32S182J	1,800	J	S	0110
RL32S202G	2,000	G	S	0111
RL32S202J	2,000	J	S	0112
RL32S222G	2,200	G	S	0113
RL32S222J	2,200	J	S	0114
RL32S242G	2,400	G	S	0115
RL32S242J	2,400	J	S	0116
RL32S272G	2,700	G	S	0117
RL32S272J	2,700	J	S	0118
RL32S302G	3,000	G	S	0119
RL32S302J	3,000	J	S	0120
RL32S332G	3,300	G	S	0121
RL32S332J	3,300	J	S	0122
RL32S362G	3,600	G	S	0123
RL32S362J	3,600	J	S	0124
RL32S392G	3,900	G	S	0125
RL32S392J	3,900	J	S	0126
RL32S432G	4,300	G	S	0127
RL32S432J	4,300	J	S	0128
RL32S472G	4,700	G	S	0129
RL32S472J	4,700	J	S	0130
RL32S512G	5,100	G	S	0131
RL32S512J	5,100	J	S	0132
RL32S562G	5,600	G	S	0133
RL32S562J	5,600	J	S	0134
RL32S622G	6,200	G	S	0135
RL32S622J	6,200	J	S	0136
RL32S682G	6,800	G	S	0137
RL32S682J	6,800	J	S	0138
RL32S752G	7,500	G	S	0139
RL32S752J	7,500	J	S	0140
RL32S822G	8,200	G	S	0141
RL32S822J	8,200	J	S	0142
RL32S912G	9,100	G	S	0143
RL32S912J	9,100	J	S	0144
RL32S103G	10,000	G	S	0145
RL32S103J	10,000	J	S	0146
RL32S113G	11,000	G	S	0147
RL32S113J	11,000	J	S	0148
RL32S123G	12,000	G	S	0149
RL32S123J	12,000	J	S	0150

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TABLE I. Part number designation - Continued

Type designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.
	Ohms			
RL32S133G	13,000	G	S	0151
RL32S133J	13,000	J	S	0152
RL32S153G	15,000	G	S	0153
RL32S153J	15,000	J	S	0154
RL32S163G	16,000	G	S	0155
RL32S163J	16,000	J	S	0156
RL32S183G	18,000	G	S	0157
RL32S183J	18,000	J	S	0158
RL32S203G	20,000	G	S	0159
RL32S203J	20,000	J	S	0160
RL32S223G	22,000	G	S	0161
RL32S223J	22,000	J	S	0162
RL32S243G	24,000	G	S	0163
RL32S243J	24,000	J	S	0164
RL32S273G	27,000	G	S	0165
RL32S273J	27,000	J	S	0166
RL32S303G	30,000	G	S	0167
RL32S303J	30,000	J	S	0168
RL32S333G	33,000	G	S	0169
RL32S333J	33,000	J	S	0170
RL32S363G	36,000	G	S	0171
RL32S363J	36,000	J	S	0172
RL32S393G	39,000	G	S	0173
RL32S393J	39,000	J	S	0174
RL32S433G	43,000	G	S	0175
RL32S433J	43,000	J	S	0176
RL32S473G	47,000	G	S	0177
RL32S473J	47,000	J	S	0178
RL32S513G	51,000	G	S	0179
RL32S513J	51,000	J	S	0180
RL32S563G	56,000	G	S	0181
RL32S563J	56,000	J	S	0182
RL32S623G	62,000	G	S	0183
RL32S623J	62,000	J	S	0184
RL32S683G	68,000	G	S	0185
RL32S683J	68,000	J	S	0186
RL32S753G	75,000	G	S	0187
RL32S753J	75,000	J	S	0188
RL32S823G	82,000	G	S	0189
RL32S823J	82,000	J	S	0190
RL32S913G	91,000	G	S	0191
RL32S913J	91,000	J	S	0192
	Megohm			
RL32S104G	0.10	G	S	0193
RL32S104J	0.10	J	S	0194
RL32S114G	0.11	G	S	0195
RL32S114J	0.11	J	S	0196

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TABLE I. Part number designation - Continued

Type designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.
	Megohm			
RL32S124G	0.12	G	S	0197
RL32S124J	0.12	J	S	0198
RL32S134G	0.13	G	S	0199
RL32S134J	0.13	J	S	0200
RL32S154G	0.15	G	S	0201
RL32S154J	0.15	J	S	0202
RL32S164G	0.16	G	S	0203
RL32S164J	0.16	J	S	0204
RL32S184G	0.18	G	S	0205
RL32S184J	0.18	J	S	0206
RL32S204G	0.20	G	S	0207
RL32S204J	0.20	J	S	0208
RL32S224G	0.22	G	S	0209
RL32S224J	0.22	J	S	0210
RL32S244G	0.24	G	S	0211
RL32S244J	0.24	J	S	0212
RL32S274G	0.27	G	S	0213
RL32S274J	0.27	J	S	0214
RL32S304G	0.30	G	S	0215
RL32S304J	0.30	J	S	0216
RL32S334G	0.33	G	S	0217
RL32S334J	0.33	J	S	0218
RL32S364G	0.36	G	S	0219
RL32S364J	0.36	J	S	0220
RL32S394G	0.39	G	S	0221
RL32S394J	0.39	J	S	0222
RL32S434G	0.43	G	S	0223
RL32S434J	0.43	J	S	0224
RL32S474G	0.47	G	S	0225
RL32S474J	0.47	J	S	0226
RL32S514G	0.51	G	S	0227
RL32S514J	0.51	J	S	0228
RL32S564G	0.56	G	S	0229
RL32S564J	0.56	J	S	0230
RL32S624G	0.62	G	S	0231
RL32S624J	0.62	J	S	0232
RL32S684G	0.68	G	S	0233
RL32S684J	0.68	J	S	0234
RL32S754G	0.75	G	S	0235
RL32S754J	0.75	J	S	0236
RL32S824G	0.82	G	S	0237
RL32S824J	0.82	J	S	0238
RL32S914G	0.91	G	S	0239
RL32S914J	0.91	J	S	0240
RL32S105G	1.0	G	S	0241
RL32S105J	1.0	J	S	0242

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Maximum continuous working voltage: 500 volts.

Power rating: 1 watt at 70°C.

Temperature cycling: Method 102 of MIL-STD-202, test condition D.
Change in resistance shall not exceed ± (1 percent + 0.05 ohm).

Low temperature operation: Change in resistance shall not exceed ± (0.5 percent + 0.05 ohm).

Short time overload: Change in resistance shall not exceed ± (0.5 percent + 0.05 ohm).

Terminal strength: Method 211 of MIL-STD-202.
Test condition A - Pull test 5 pounds.
Test condition D - Twist test: rotated 360° for 3 rotations. Change in resistance shall not exceed ± (0.5 percent + 0.05 ohm).

Dielectric withstanding voltage: Methods 301 and 105 of MIL-STD-202.
Atmospheric pressure - 1,000 volts rms.
Barometric pressure - 500 volts rms.

Insulation resistance: Method 302 of MIL-STD-202. Test condition A or B, whichever is practicable - Insulation resistance shall be not less than 100 megohms.

Resistance to soldering heat: Method 210 of MIL-STD-202.
Test condition A - Change in resistance shall not exceed ± (0.5 percent + 0.05 ohm).

Moisture resistance: Method 106 of MIL-STD-202. Change in resistance shall not exceed 1.5 percent. Insulation resistance shall be 100 megohms, minimum.

Resistance temperature characteristic: Method 304 of MIL-STD-202. Change in resistance shall not exceed ± 0.020 percent per degree C.

Life: Method 108 of MIL-STD-202.
Test condition D - Change in resistance shall not exceed 2.0 percent.

Shock, medium impact: Method 205 of MIL-STD-202.
Test condition C - Change in resistance shall not exceed ± (0.5 percent + 0.05 ohm).

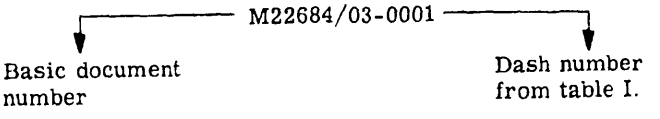
Vibration, high frequency: Method 204 of MIL-STD-202.
Test condition D - Change in resistance shall not exceed ± (0.5 percent + 0.05 ohm).

Solderability: Method 208 of MIL-STD-202.

Ⓑ Substitution data. The resistors specified herein are not for use in design after date of this specification sheet. They are suitable for use in design contracts current to that date and to support existing military equipment. Resistors specified in MIL-R-39017/3 are preferred for design and regardless of failure rate can be used as substitutes for the inactivated resistors of the same resistance value and tolerance of this specification.

Part number: Consists of the basic number of this specification sheet and a dash number taken from table I.

Example:



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Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - EC, AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC, CG
Air Force - 19

Preparing activity:
Army - EL

(Project 5905-0723)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-R-22684/3B, RESISTORS, FIXED, FILM, INSULATED, STYLE RL32

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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